LABORATORY 2: MODEL BASED CONTROL

31 oktober 2023 (version: C19)

1 Introduction

Laboratory 1 gave an introduction to the principles of feedback control and, in particular, PID control. The laboratory also gave some experience in the difficulties involved in tuning controllers based on trial and error. Laboratory 2 will show how feedback controllers that provide specific system properties can be derived in a systematic fashion. The derivation is based on a two-step procedure. First a model of the process is constructed based on a combination of physical insight and experimental data. The model is then used to construct a feedback control law that yields certain desired properties in the closed-loop system. Pole placement design and frequency domain loop shaping (i.e., lead-lag compensation) will be considered. For the two-tank process, the resulting control law will with either approach be possible to implement in the form of a PID controller.

The process

The laboratory investigation will be performed on the same tank system that was used in Laboratory 1 (see Figure 1).

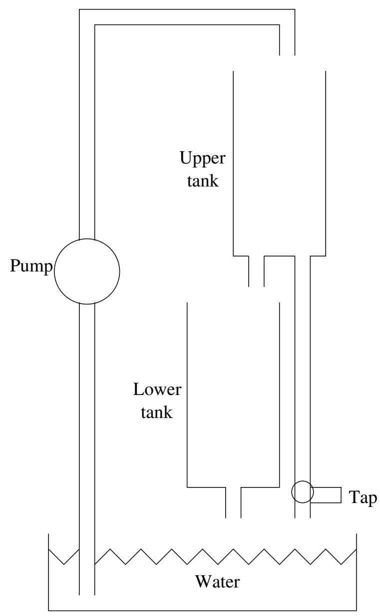

Figure 1: Coupled tanks with pump.

Preparations

Laboratory 1 must have been completed before conducting Laboratory 2. If this is not the case, contact the teacher before coming to the laboratory.

Before going to the laboratory, you need to prepare as listed below. Document your preparations in the manual, and bring to the laboratory. *You will not be allowed to complete Laboratory 2 without preparations.*

- You need to pass the Pre-Lab2 Quiz in Canvas.
- Preparation tasks 2.1-2.7, 4.1, 4.2 and 5.1 in this manual should be solved.
- Perform all other tasks in this manual using the MATLAB simulation software. Document your preliminary results in the manual. The assistant will then ask you to complete and report parts of the tasks in the laboratory. Note in particular deviations between the simulated and physical process, and discuss causes and the other tasks with the assistant.
- Briefly explain the following concepts:

Linearization:

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Poles and zeros:

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Frequency response:

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Crossover frequency:

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Phase margin:

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Bandwidth:

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Loopshaping, or lead-lag compensation:

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2 Modelling

2.1 Physical model of the process

Task 2.1 (*Preparation task)*
The relationship between the outflow velocity $v(t)$ [m/s] and the water level $h(t)$ [m] in a tank is given by *Bernoulli's law*

$$v(t) = \sqrt{2gh(t)}$$

Write down the differential equations for the tank system in Figure 1 when the dynamics in the motor and the hoses are neglected. Let A_1 and A_2 be the cross-sectional areas of the tanks and a_1 and a_2 the effective outlet areas. Assume that the flow q from the pump is directly proportional to the voltage u applied to the motor.

Task 2.2 (*Preparation task)*
Assume that both tanks have the same cross-sectional area, i.e., $A_1 = A_2 = A$, and show that the model then can be written

$$\begin{aligned} \frac{dh_1(t)}{dt} &= -\alpha_1\sqrt{2gh_1(t)} + \beta u(t) \\ \frac{dh_2(t)}{dt} &= \quad \alpha_1\sqrt{2gh_1(t)} - \alpha_2\sqrt{2gh_2(t)} \end{aligned} \tag{1}$$

where $\alpha_1 = a_1/A$ and $\alpha_2 = a_2/A$. Note that the model now only contains the three parameters α_1, α_2 and β.

Task 2.3 (*Preparation task*)
Determine the parameters α_1, α_2 and β from the following construction data:

Diameter for the outlet holes:	d	$= 4.763 \cdot 10^{-3} m$
Diameter for the tanks:	D	$= 44.45 \cdot 10^{-3} m$

A pump voltage of $11V$ gives a flow of $50 \cdot 10^{-6} m^3/s$

Task 2.4 (*Preparation task*)
Determine the relationship between the levels in the upper and lower tank at steady-state conditions, i.e., determine

$$\gamma = \frac{h_2^0}{h_1^0},$$

for an equilibrium point

$$h_1 = h_1^0,\ h_2 = h_2^0,\ u = u^0. \qquad (2)$$

Task 2.5 (*Preparation task*)
Introduce the scaled deviation variables

$$\begin{aligned}
\Delta x_1(t) &= 400[h_1(t) - h_1^0] \\
\Delta x_2(t) &= 400[h_2(t) - h_2^0] \\
\Delta u(t) &= \frac{100}{15}[u(t) - u^0]
\end{aligned}$$

and linearize the system (1) around the equilibrium point (2). Note that the levels and the input have been scaled so that they are between 0 and 100.
Hint: Start by linearizing the system and then introduce the new variables $\Delta x_1(t)$, $\Delta x_2(t)$, *and* $\Delta u(t)$.

Task 2.6a (*Preparation task*)
Show that the linearized system can be described with the following transfer functions

$$\Delta X_1(s) = \frac{k}{1+\tau s}\Delta U(s)$$
$$\Delta X_2(s) = \frac{\gamma}{1+\gamma\tau s}\Delta X_1(s)$$

where $\tau = \frac{1}{\alpha_1}\sqrt{\frac{2h_1^0}{g}}$ and $k = 60\beta\tau$.

Task 2.6b (*Preparation task*)
Calculate the values of k, τ and γ when h_2^0=0.1 m.

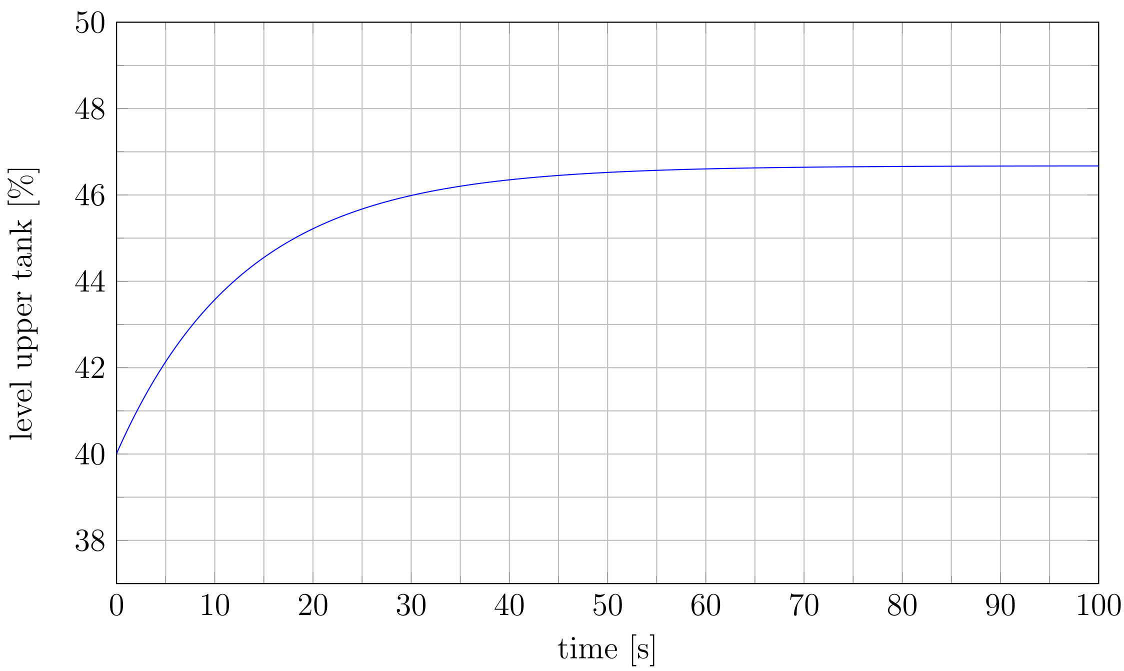
Task 2.7 (*Preparation task*)
Assume that the process is steady state at $t = 0$, at which a step is made in the input signal with $\Delta u = 4$ scale units. The linearized model then gives

$$\Delta X_1(s) = \frac{k}{1+\tau s}\frac{4}{s} \quad \Rightarrow$$
$$\Delta x_1(t) = 4k(1 - e^{-t/\tau})$$

When the system has settled in a new steady state, the level in the upper tank has changed with $4k$ units. At $t = \tau$ the change is

$$\Delta x_1(\tau) = 4k(1 - e^{-1}) = 0.632 \cdot 4k$$

Determine k and τ from the step response in the figure below.



3 Experimental modeling

In the previous section, we derived a mathematical model of the tank system based on physical insight. In this section, we will determine the parameters in that model based on experiments.

Task 3.1
Start the program by double clicking the Matlab icon on the desktop. The GUI is the same as used in Laboratory 1.

- Turn the **Anti windup** switch to **On**.
- Display both tank levels by selecting them under **Plot settings** and **Signals**.
- Manually change the level in the lower tank to 40 ± 1. *This means that you have to find the input signal that gives this level in the lower tank.*
- Wait until the process is in steady state and write down the input value: $u_0 =$.......
- Determine the γ-value for the process be calculating the ratio between the levels in the lower and upper tank: $\gamma =$.......

Task 3.2
Note: In this task you should look at the upper tank!

- Change the input signal to $u_0 + 4$.
- Wait 100 seconds, and then use the **Capture** button to capture the step response.
- Determine k and τ in the same way as you did in task 2.7. Use the zoom-function in the capture-figure to improve the accuracy of your readings. $k =$....... $\tau =$.......
- Check with one of the assistants that the obtained values are reasonable.

Task 3.3
Are k, τ and γ equal to the ones that you derived in task 2.6b? Discuss the reasons for possible differences!

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4 Pole placement design

We have now derived a model of the system. This model shall next be used to construct a controller. The model-based controller design methods that we will study in this laboratory are pole placement design and loop shaping, or lead-lag design.

We will first study pole placement design for control of the level in the upper tank. Note from the model that the transfer function from ΔU to ΔX_1 is a first order system. Feedback control with a PI-controller, which also has one pole, then results in a closed-loop system with two poles. Remember that the PI-controller, in parallel form, is given by

$$F(s) = K \left[1 + \frac{1}{T_I s} \right]$$

Since we have two free controller parameters, K and T_I, we can place the two poles anywhere in the complex plane, with the only restriction that they must be conjugate when containing an imaginary part.

Task 4.1 (*Preparation task*)
Show that, for a first order system $G(s) = k/(\tau s + 1)$, a pole placement at $s = -\beta \pm i\beta$ corresponds to a PI-controller with parameters

$$K = (2\beta\tau - 1)/k \qquad\qquad T_I = Kk/(2\beta^2\tau)$$

Note that we with this specific pole placement have restricted the closed loop poles to lie on the bisectors of the 3rd and 4th quadrant.

Task 4.2 (*Preparation task*)

We shall now study pole placement design for control of the level in the lower tank.

Note from the model that the transfer function from ΔU to ΔX_2 is a second order system. A PI-controller would thus yield three poles in the closed loop system, but with only two free controller parameters. To be able to place all poles we therefore need to employ a PID-controller.

The ideal PID-controller, in parallel form, is given by

$$F(s) = K\left[1 + \frac{1}{T_I s} + T_D s\right]$$

However, this controller is not suitable for implementation because it is not proper, i.e., it contains more zeros than poles, and furthermore, potentially highly sensitive to measurement noise due to the differentiation of the control error. To overcome these problems, commercial PID-controllers usually contain a filter for the D-part. In this laboratory, the PID-controller with filter is implemented as

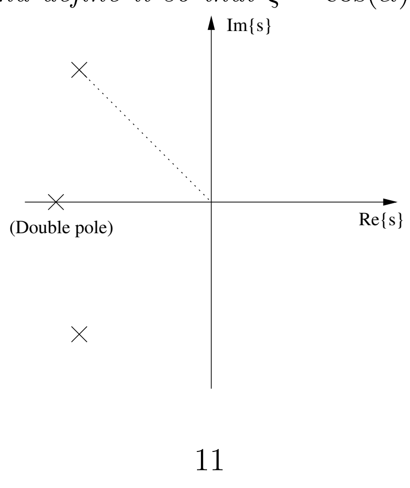
$$F(s) = K\left[1 + \frac{1}{T_I s} + \frac{T_D N s}{s + N}\right]$$

This controller has two poles, and the closed-loop system for the level in the lower tank will hence contain four poles. Note that we now also have four parameters (K, T_I, T_D, N) so that all four poles can be placed arbitrarily in the complex plane.

We will here consider a design in which two poles are forced to be real and equal, corresponding to the closed-loop characteristic equation

$$(s + \chi)^2(s^2 + 2\zeta\omega_0 s + \omega_0^2) = 0$$

Show in the figure below what the parameters χ, ζ and ω_o correspond to.
Hint: Introduce the angle α and define it so that $\zeta = \cos(\alpha)$.



Task 4.3

In order to study how the choice of the poles affect the behavior of the closed-loop system, we will place the double real pole at $-\chi$ relatively far away from the origin compared to the two complex conjugate poles. Since poles far away from the origin are faster than poles closer to the origin, this means that the two complex poles will be the dominant ones.

1. Under **Controller settings**, select **PID** and **Tank2** to be controlled.
2. To tune the PID parameters, click **Tuner**.
3. Select the **Pole placement** tab and enter the system parameters; k, τ and γ. (Note that if you close the tuner, you will have to re-enter these parameters.)
4. Choose χ=0.5, ζ=0.7, and ω_0=0.1.
5. Click **Save** under **Controller settings** on the main GUI to transform and transfer the corresponding PID parameters.

Set the reference value to 40 and wait until the system is steady state. Change the reference from 40 to 45. Write down the rise time T_r, the overshoot M, and the settling time T_s in the table below.

After you complete it, change the reference back to 40. Wait until the system is steady state and then repeat the experiment for the other pole placements given in the table below.

χ	ζ	ω_0	T_r	M	T_s
0.5	0.7	0.1			
0.5	0.7	0.2			
0.5	0.1	0.2			
0.5	1.0	0.2			

How does M, T_r and T_s, respectively, depend on ζ and ω_0? Compare with what you would expect from theory.

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What would happen if you chose ζ=0?

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Does the overshoot obtained with χ=0.5, ζ=1.0, and ω_0=0.2 correspond to what you would expect from theory? Can you explain the result that you get?
Hint: consider the closed-loop zeros.

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In pole placement design, we limit ourselves to consider the location of the system poles only. In many cases this will yield acceptable results. However, as seen above, the zeros of the closed-loop can in some cases also have a significant impact on the closed-loop response. In these cases, a design method which takes a more general approach to the system properties is preferred. One such approach is shaping of the systems frequency response, through frequency domain loop shaping (lead-lag compensation), which is considered next.

If any of the assistants are available, then this is a good opportunity for partial reporting of your results. Otherwise, continue working and report later.

Partial report accepted by:

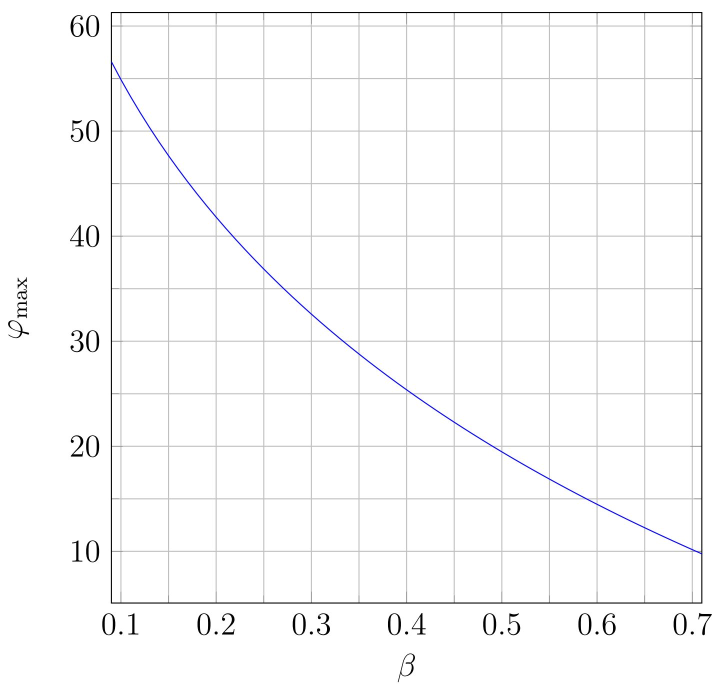
5 Frequency domain loop shaping

The frequency domain is well suited for systematic model-based design of controllers in order to achieve desired system properties such as speed of response, damping, and steady-state control error. In the frequency domain these properties are represented by the bandwidth ω_B, resonance peak M_p, and steady-state gain of the closed-loop system $G_c(0)$, respectively. Since the relationship between the controller and the open-loop system properties is more transparent than for the corresponding closed-loop system, we usually prefer to shape the open-loop frequency response. The related properties for the frequency response of the open-loop system $G_o(s) = F(s)G(s)$ are the crossover frequency ω_c, the phase margin φ_m, and steady-state gain $G_o(0)$, respectively.

To shape the open-loop frequency response, we will here employ a lead-lag controller with single lead and lag elements

$$F(s) = K \underbrace{\frac{\tau_D s + 1}{\beta \tau_D s + 1}}_{F_{lead}} \underbrace{\frac{\tau_I s + 1}{\tau_I s + \gamma}}_{F_{lag}}.$$

The lead-element F_{lead} is used to "lift" the phase so as to obtain a desired phase margin φ_m. The maximum phase lift as a function of β is shown in Figure 5.1. The lag-element F_{lag} is used to increase the loop-gain at low frequencies in order to reduce the steady-state control error. If a zero steady-state error is required, then $\gamma = 0$. The gain K is used to provide the desired crossover frequency ω_{cd}.



Task 5.1 (*Preparation task*)
The lead-lag controller

$$F(s) = K\frac{\tau_D s + 1}{\beta\tau_D s + 1}\frac{\tau_I s + 1}{\tau_I s + \gamma}$$

with $\gamma = 0$, can be implemented with a PID-controller on parallel form

$$F_{pid}(s) = K_{pid}\left[1 + \frac{1}{T_I s} + \frac{T_D N_{pid} s}{s + N_{pid}}\right]$$

Write both controllers on the normal form

$$F(s) = \frac{as^2 + bs + c}{s^2 + ds}$$

and derive the equations that relate the parameters so as to make the two above controllers equal.
Note: You are not expected to solve the system of equations.

Task 5.2 We shall now employ loopshaping, or lead-lag design to derive a controller for the level in the lower tank.

First study the Bode diagram for the uncompensated system G, which can be generated through the following steps

1. Click on **Tuner**, and then on the tab **Lead lag**. *Note that the control law is implemented as a PID.*
2. Enter the system parameters; k, τ and γ. (Note that if you close the tuner, you will have to re-enter these parameters.)
3. Plot the Bode diagram for G.

Determine the crossover frequency ω_c and the phase marginal φ_m for the uncompensated system G. $\omega_c =$ ...... $\varphi_m =$ ......

Task 5.3 Now use the controller F to design an open-loop system FG with crossover frequency ω_{cd} =0.2 rad/sec and phase margin $\varphi_m = 70°$. This can be done as described below

1. Calculate the required phase increase at ω_{cd} to achieve the desired phase margin. (Recall that the lag-part will decrease the phase with approx. 5.7°.) Required phase increase = ...
2. Determine the β-value that gives this phase increase (see figure 5.1).
$\beta =$ ....
3. Choose the parameter τ_D such that the maximum phase lift is obtained at the desired crossover frequency ω_{cd}: $\tau_D = \frac{1}{\omega_{cd}\sqrt{\beta}}$ ....
4. Choose the parameter $\tau_I = 10/\omega_{cd} =$ ....
5. Plot the bode diagram for L=FG with controller gain $K = 1$ and read the amplitude $|FG|_{dB}$ in decibel at ω_{cd}. Calculate the corresponding amplitude $|FG|$=$10^{|FG|_{dB}/20}$. (An example: if $|FG|_{dB} = -12$ then $|FG| = 10^{-12/20}$. Note the minus sign!) Determine the controller gain K that gives an amplitude of 1 at the desired crossover frequency, ω_{cd}. $K =$ ....
6. The corresponding closed-loop Bode diagram may be plotted by clicking G_c.
7. Plot the Bode diagram for $L = FG$ with the determined controller gain K and verify that you have obtained the desired frequency response.
8. Click on **Save** in the main GUI to calculate and implement the corresponding PID parameters.

We will consider the performance of the obtained controller by performing experiments similar to those performed for the pole placement design above.

Complete the table below with the closed-loop Bode plot variables: the bandwidth frequency ω_B and the resonance peak M_p (in dB).

Then, set the reference value to 40 and wait until the system is steady state. Change the reference from 40 to 45. Write down the rise time T_r, the overshoot M, and the settling time T_s in the table below.

Finally, change the reference back to 40. Wait until the system is in steady state and repeat the experiment for the other combinations listed in the table below.

ω_c	φ_m	ω_B	M_p	T_r	M	T_s
0.2	70					
0.2	60					
0.2	50					
0.4	60					
0.8	60					

How are the closed-loop variables ω_B and M_p related to the open-loop variables ω_c and φ_m? (Hint: use Figures 5.11 and 5.12 on page 104 in Glad&Ljung.)

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How do T_r, M and T_s depend on ω_c and φ_m? Compare with what you would expect from theory. (Hint: use Figures 5.11 and 5.12 on page 104 in Glad&Ljung.)

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6 Summary

Task 6.1

- What are the advantages and disadvantages with model based control design compared to trial-and-error tuning of a PID controller.

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- Summarize the results you found with pole placement design in this laboratory and compare with what you would expect from theory.

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- Summarize the results you found with lead-lag design and compare with what you would expect from theory.

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